

Frédéric Andrian

Junior Software Developer · C#, Python and Full-Stack Applications

Bordeaux, France

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Professional Summary

Junior Software Developer completing an MSc-equivalent engineering degree in Artificial Intelligence and Cognitive Science. At INRIA (French National Institute for Research in Digital Science and Technology), I independently delivered an end-to-end web application from a user need to a functional solution, including Python and FastAPI services, REST APIs, MongoDB, a React and Next.js interface, Dockerized components, and GitHub CI/CD. During my studies, I also developed software projects using C#, .NET, Entity Framework, and Java. Curious and autonomous, I want to keep learning, contribute throughout the software lifecycle, and grow within Mendix's international engineering environment. Available from November 2026.

Experience

INRIA (French National Institute for Research in Digital Science and Technology), Mnemosyne Team

Bordeaux, France

AI Research Engineer Intern

Mar. 2026 – Sep. 2026

- Independently designed and developed, with guidance from my supervisors, an end-to-end web application that helps researchers discover scientific papers.
- Built Python and FastAPI backend services, REST APIs documented with OpenAPI, and MongoDB persistence.
- Developed a React and Next.js interface that enabled researchers to interact with the application and its tools.
- Integrated the OpenAI API and designed the tools used by the agent to retrieve information and complete successive tasks.
- Built the agent without an orchestration framework to understand and control tool execution, state, memory, context, and multi-step behavior.
- Implemented a RAG pipeline using document chunking, embeddings, semantic search, and ChromaDB.
- Dockerized selected components and implemented a GitHub CI/CD pipeline to automate integration.
- Managed a Dockerized web application for LLMs installed locally on a Mac Studio, including model installation and configuration.
- Developed a Python tool to automatically collect scientific paper metadata from multiple public sources.
- Implemented webhooks to relay the results of scheduled tasks.
- Evaluated OpenClaw in an isolated Docker environment with attention to the reliability and security of autonomous tool and file access.
- Benchmarked Qwen, Gemma, and GLM on an HPC cluster with Slurm across latency, TTFT, response quality, and context size.
- Evaluated multiple quantization configurations to identify the best trade-off between inference performance and response quality for each execution environment.
- Defined an experimental protocol, analyzed the results, and contributed to a scientific manuscript documenting the model evaluation.

Karunya University

Coimbatore, India

Reinforcement Learning Research Intern

Jun. 2025 – Sep. 2025

- Designed and compared deterministic path planning with a reinforcement learning approach for autonomous navigation of a hospital robot.
- Deployed and optimized the software and reinforcement learning model on an embedded Raspberry Pi.

Academic Project

University Management REST API

ENSC, Bordeaux INP

C#, ASP.NET Core, Entity Framework Core, SQLite

2024

- Developed a REST API in C# and ASP.NET Core to manage university student and course data.
- Implemented CRUD endpoints with controllers and DTOs, using Entity Framework Core and SQLite for data modeling, persistence, and migrations.
- Added Swagger/OpenAPI documentation to explore and interact with the API.

Technical Skills

Programming	C#, .NET, ASP.NET Core, Entity Framework Core, Python, Java, Node.js
Backend and APIs	FastAPI, REST APIs, Swagger/OpenAPI, MongoDB, SQLite, SQL, webhooks
Frontend	React, Next.js
Software Engineering	Linux, Docker, Git/GitHub, CI/CD, API design, full-stack development
AI-Assisted Development	ChatGPT
AI Systems	LLMs, OpenAI API, multi-step agents, tool design, RAG, PyTorch, Deep Learning
Model Evaluation	Qwen, Gemma, GLM, quantization, experimental protocols, latency, TTFT, response quality
Languages	French (native), English (professional, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP

Bordeaux, France

Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science

2023 – 2026